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Machine information

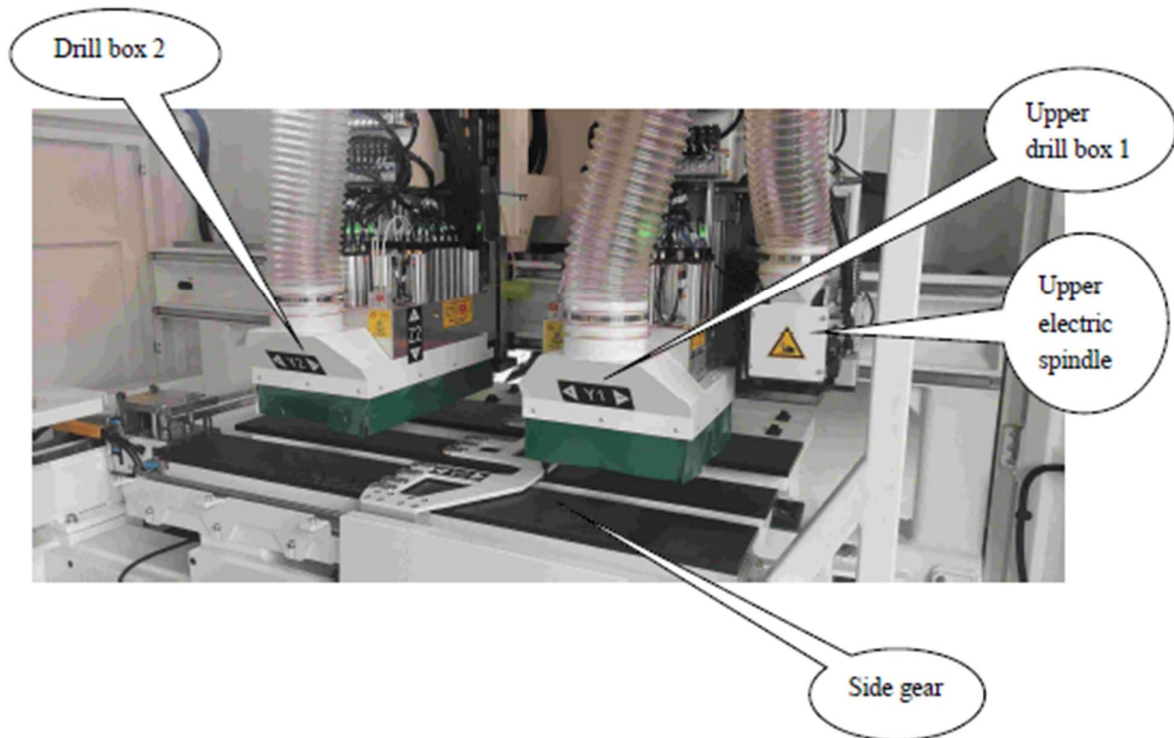
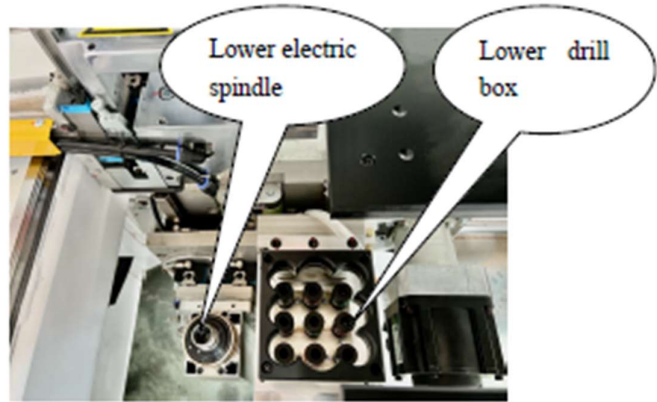
Model		NCB612S			
		Stand-alone machine			
Sheet width		30-1200mm			
Length of sheet		200-3000 mm			
Thickness of sheet		10-60 mm			
Maximum speed	X-axis, U-axis	135 m/min			
	Y1-axis, Y2-axis, V-axis	75 m/min			
	Z1-axis, Z2-axis, W-axis	30 m/min			
Gang drill set	Number of vertical drills on the upper machine head*	18			
	Number of horizontal drills on the upper v head	10			
	Number of vertical drills on the lower machine head	9			
	Speed	5000 rpm			
Electric spindle	Power	3.5kW (upper)+3.5kW (lower)			
	Tool installation (tool tip to chuck plane) mm	≤35			
	Speed	18000 rpm			
Saw blade motor	Power	1.7KW			
	Saw blade	100×4×22×20			
	Speed	2800 rpm			
Total power		19.73 kW			
Overall dimensions (L x W x H)		6150×2273×2325 mm			
Weight		3600 kg			

The main technical parameters of the machine are as follows:

Model		NCB612	NCB612DX	NCB612DX T	NCB612D XL	NCB612D XTL
		Stand-alone machine	Stand-alone machine	Stand-alone machine	Stand-alone machine	
Sheet width		30-1200mm	30-1200mm	30-1200 mm	50-1200 mm	50-1200 mm
Length of sheet		200-3000 mm	200-3000 mm	200-3000 mm	250-2800 mm	200-2800 mm
Thickness of sheet		10-60 mm	10-60 mm	10-60 mm	10-60 mm	10-60 mm
Maximum speed	X-axis, U-axis	135 m/min	135 m/min	135 m/min	135 m/min	135 m/min
	Y1-axis, Y2-axis, V-axis	75 m/min	75 m/min	75 m/min	75 m/min	75 m/min
	Z1-axis, Z2-axis, W-axis	30 m/min	30 m/min	30 m/min	30 m/min	30 m/min
Gang drill set	Number of vertical drills on the upper machine head*	18	2×18	2×18	2×18	2×18
	Number of horizontal drills on the upper v head	10	2×10	2×10	2×10	2×10
	Number of vertical drills on the lower machine head	9	9	9	9	9
	Speed	5000 rpm	5000 rpm	5000 rpm	5000 rpm	5000 rpm
Electric spind	Power	3.5kW (upper)+3.5 kW (lower)	3.5kW (upper)+3.5kW (lower)	5.5 kW (upper) + 3.5 kW (lower)	3.5kW (upper)+3.5k W (lower)	5.5 kW (upper) + 3.5 kW (lower)

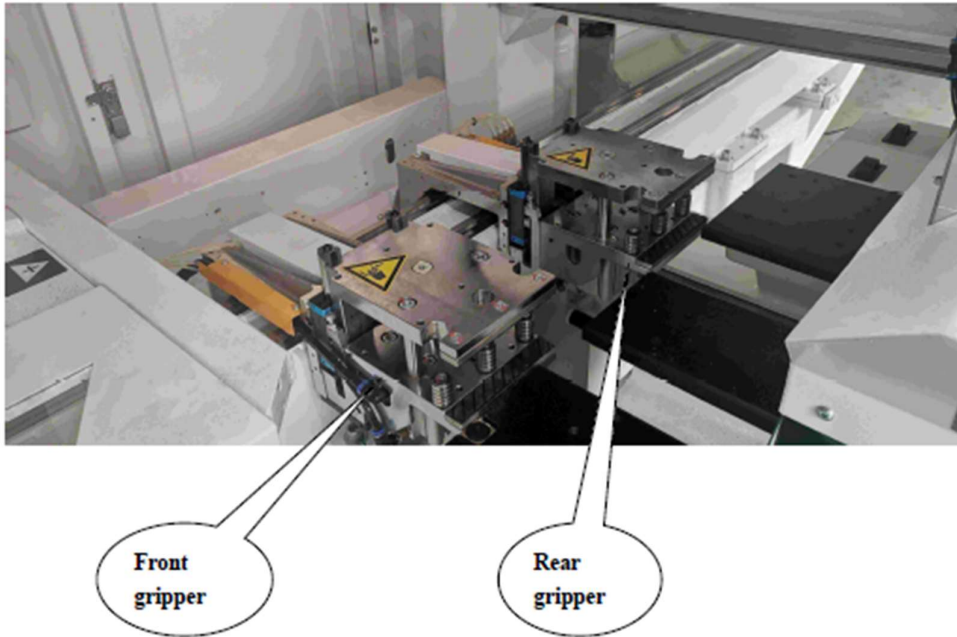
Model		NCB612	NCB612DX	NCB612DX T	NCB612D XL	NCB612D XTL
le	Speed	18000 rpm	18000 rpm	18000 rpm	18000 rpm	18000 rpm
Total power		18.03 kW	21.93KW	23.93 kW	22.68 kW	25.43 kW
Overall dimensions (L x W x H)		6150×2273 ×2325mm	6150×2806×23 25mm	6150×2806×23 25mm	7390×2806×2 325mm	7390×2806× 2325mm
Weight		3600 kg	3750 kg	3850 kg	3950 kg	4050 kg

Machine description



Double gripper

There are front and rear grippers on the working platform, which are respectively driven by a servo motor and driven by gears, racks, sliding blocks and guide rails. It can move along the longitudinal direction, which are defined as "X" axis (front gripper), "U" axis (rear gripper).



Upper Machine Head

The upper machine head is composed of left and right drill boxes and an electric spindle.

The electric spindle and the right-side drill box are mounted on the same board.

Driven by a servo motor, driven by gears, racks, sliders and guide rails; the left and right drill boxes can move horizontally and vertically respectively.

1. Lateral movement is defined as "Y1" axis (right side drill box + electric spindle), and

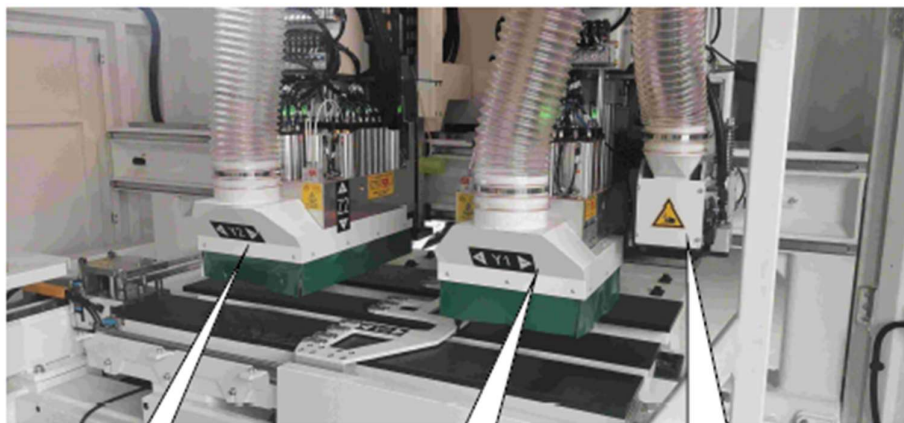
"Y2" axis (left side drill box);

2. Longitudinal movement is defined as "Z1" axis (right side drill box + electric spindle),

and "Z2" axis (left side drill box).

In addition to synchronous horizontal and vertical movement with the right-side drill box;

The upper electric spindle can also move vertically up and down independently, controlled by an air cylinder.



Drill
box 2

Drill box 1

Electric spindle

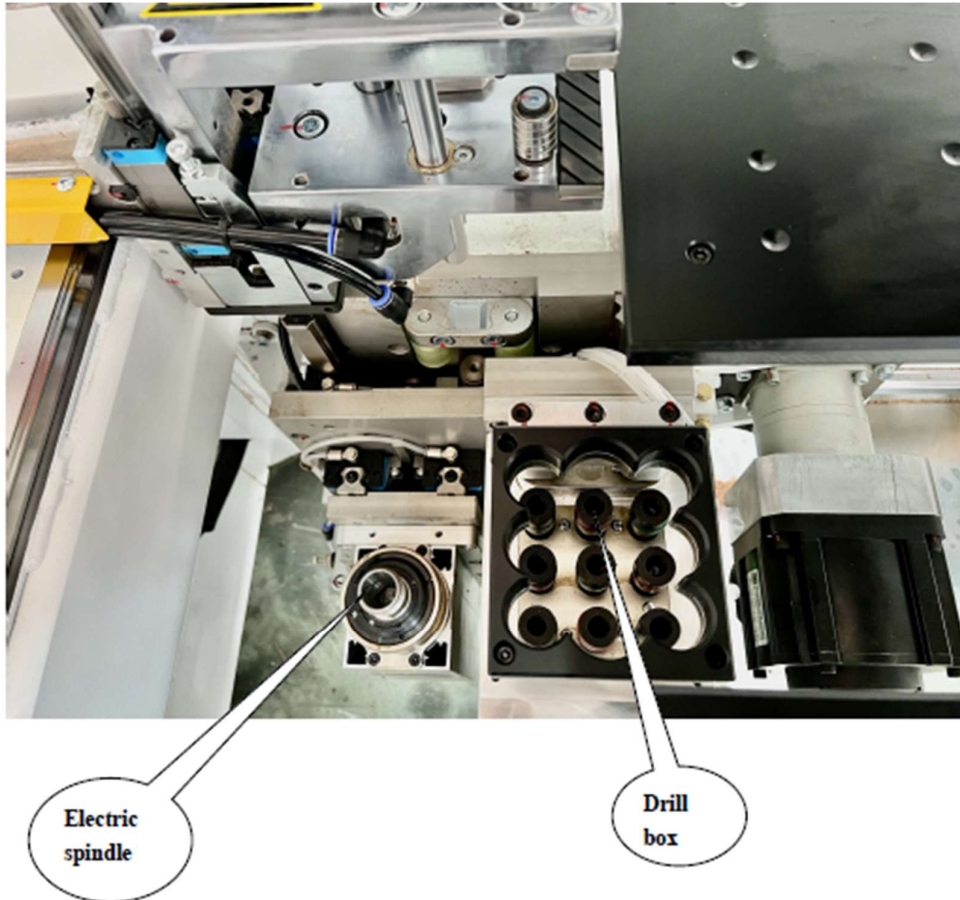
Remarks: The electric spindle shown is on the right side, and it can also be mirrored on the left side.

Lower the machine head

mounted on the same board.

Driven by servo motor, driven by gears, racks, sliders and guide rails; the lower head can move horizontally and vertically, which are defined as "W" axis and "V" axis respectively.

In addition to the synchronous lateral and vertical movement with the lower drill box, Lower electric spindle can also independently move up and down vertically through the control of the air cylinder.



Automatic side stop

The side crosspiece is driven by a servo motor, driven by gears, racks, guide rails and slide rails: it can move laterally, which is defined as the "V" axis.



Air-floating platform (Only on Manual models.)

When the board is loaded and unloaded, the air-floating platform makes the board easy to move and protects the board from scratches.

Model NCB612/NCB612DX/NCB612DXT



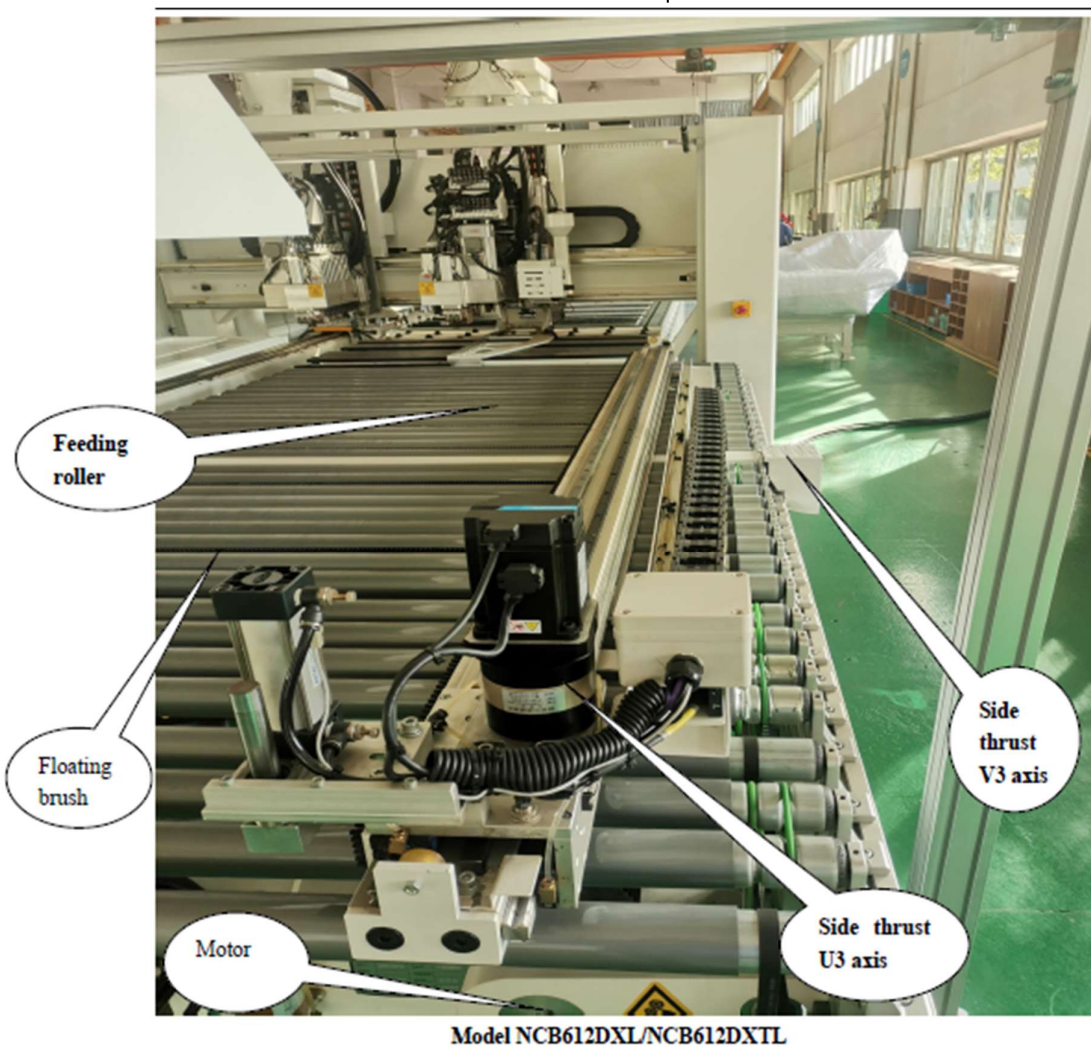
Feeding table (Only on Automatic models.)

The side push V3 is driven by a servo motor and driven by a screw rod; it can move laterally, which is defined as the "V3" axis.

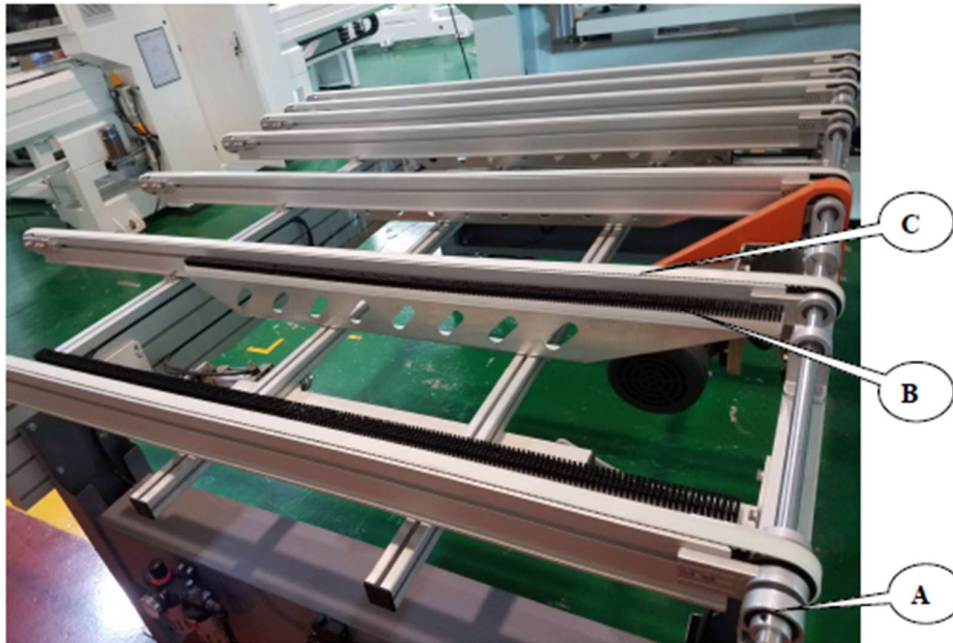
The side push U3 is driven by a servo motor, driven by gears, racks, guide rails and slide rails; it can move longitudinally, which is defined as the "U3" axis.

The brush is controlled up and down by the cylinder

The feeding roller is controlled by a motor and a frequency converter. There are three sensors under the drum, which are start, deceleration and stop.



Rear discharging conveyor (Manual model.)



Model NCB612/NCB612DX/NCB612DXT

A - Power transmission

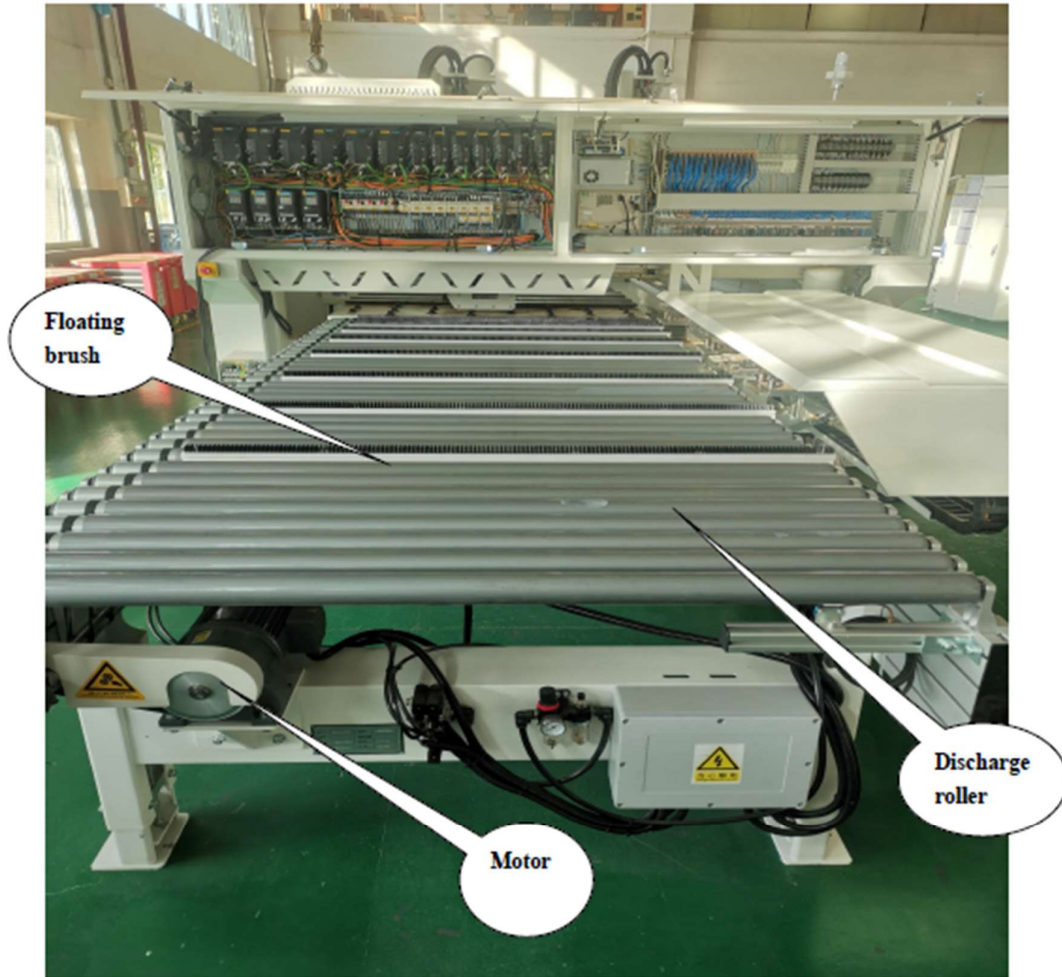
B - Floating brush

C - High strength conveyor belt

Rear discharging conveyor (Automatic model.)

The brush is controlled up and down by the cylinder

The discharge drum is controlled by the discharge motor and the frequency converter. There are three sensors under the drum, which are start, deceleration and stop.



Model NCB612DXL/NCB612DXTL

Control system

Windows based control system, with stop and start buttons on the unit.



Electric cabinet



Axes and handwheel

Definition of axes

The axes and motion units controlled by servo motors are defined as follows:

Definition of axes	Movement unit	Axial scale of handwheel*
"X" axis	Longitudinal movement of gripping unit 1 (front)	X
"Y1" axis	Lateral movement of upper drill box 1	Y
"Z1" axis	Vertical movement of upper drill box 1 (synchronized with the upper electric spindle)	Z
"U" axis	Longitudinal movement of gripper unit 2 (rear)	4
"V" axis	Vertical movement of lower head (synchronized with lower electric spindle)	5
"W" axis	Lateral movement of lower machine head (synchronized with lower electric spindle)	6
"V2" axis	Horizontal movement of the side gear unit	7
"Y2" axis	Lateral movement of upper drill box 2	8
"Z2" axis	Vertical movement of upper drill box 2	*

*Press the "Axis Switch" key on the main page of the software once to switch to the other axes (the scale is full, go to the next page).

Handwheel operation

In manual mode or handwheel simulation mode, manual operation of the machine is allowed. The movement of each axis of the machine is performed by shaking the handwheel.

There are scale selection and speed magnification selection on the handwheel.



Handwheel magnification	Actual displacement of each grid
×1	0.01mm
×10	0.1mm
×100	1mm

Transport of the machine

The transportation of the machine can be by road, rail, sea, or air. It depends on the sales contract.

The movement of the machine in the workplace must be carried out by a forklift or crane and by fully trained and qualified personnel.

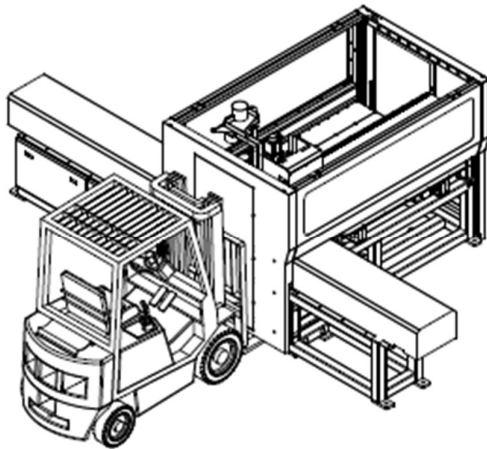
The lifting tool is in good working condition.

The load capacity of the lifting tool meets the weight of the machine (see the machine nameplate).

Do not move suddenly to avoid unbalance and falling of the machine.

The center of gravity of the machine is in the middle of the stress point of the lifting tool; avoid the machine from being unbalanced and colliding.

Avoid the machine touching the ground.



Site preparation

The machine is installed in an indoor space and must be installed on a flat concrete floor. The load capacity on the ground meets the weight of the machine (see the machine nameplate).

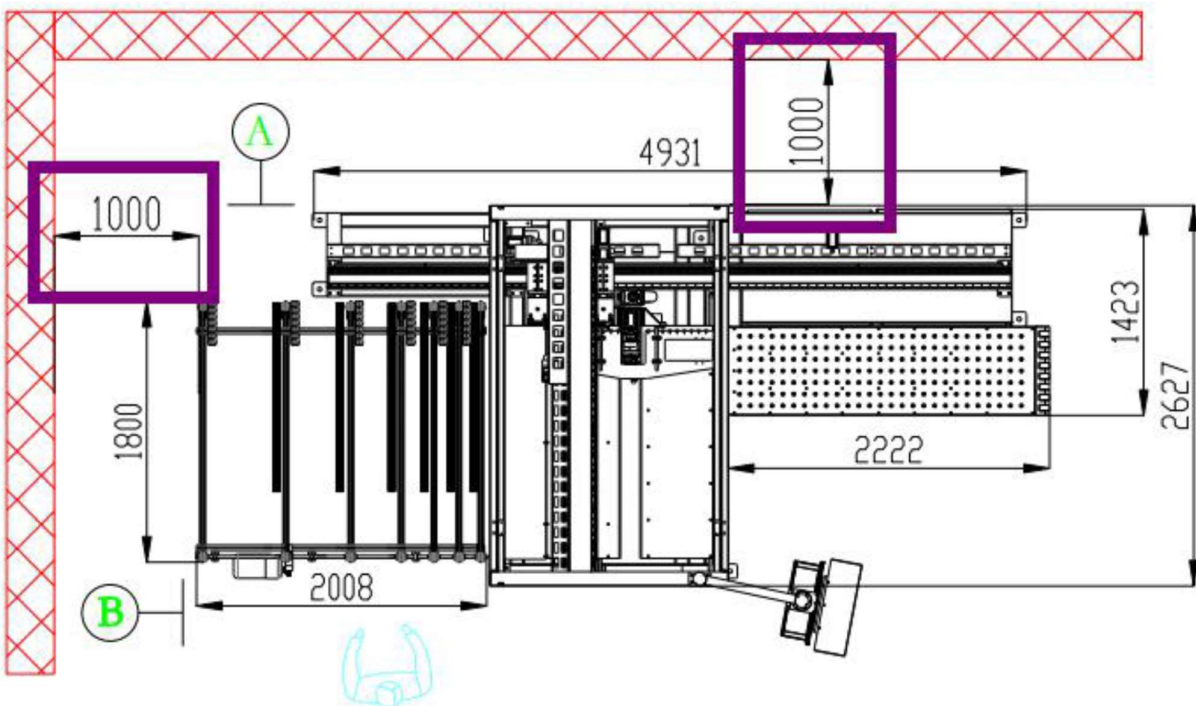
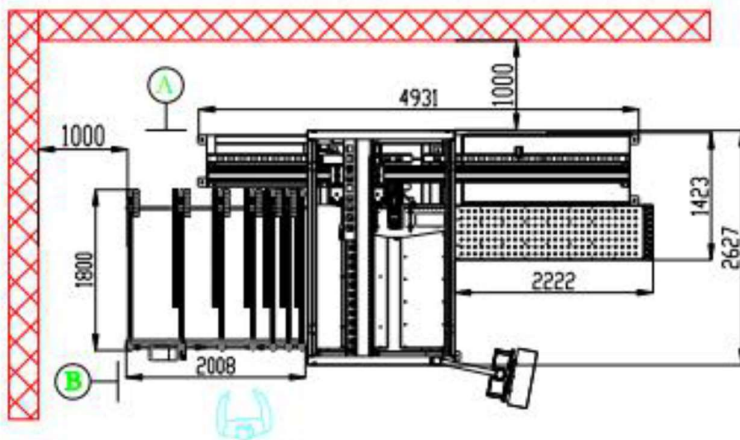
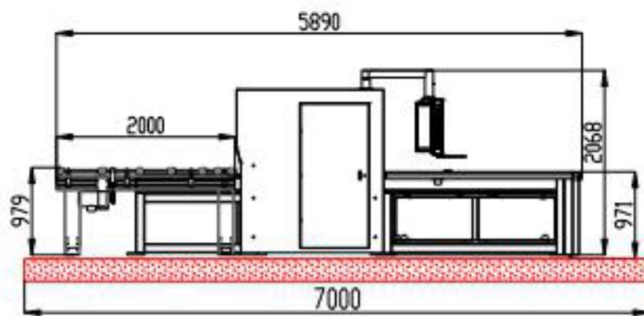
The allowable error of ground levelness is 0.5mm/m, and the maximum inclination in all directions is less than 0.3%.

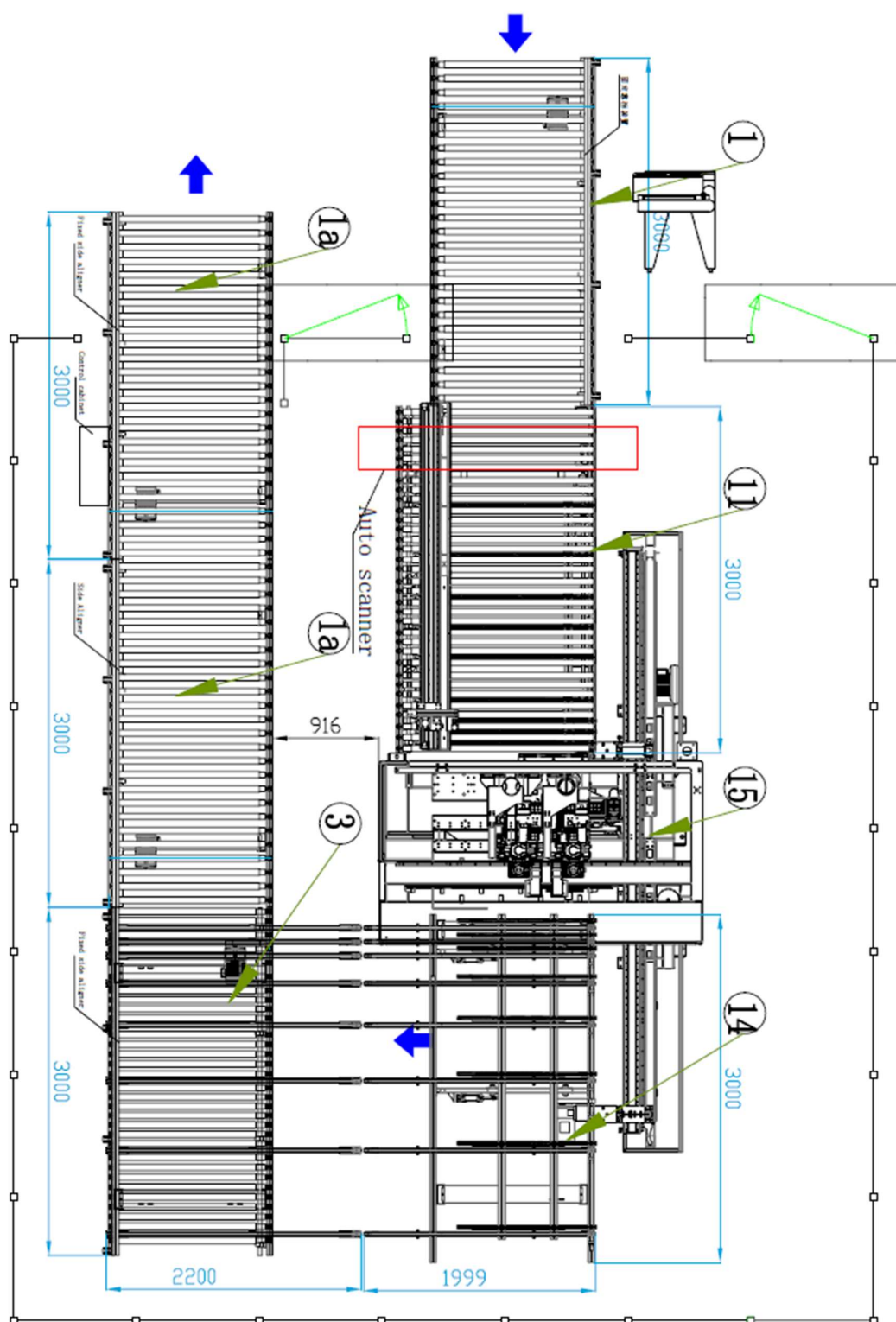
Environment of the machine installation area:

Temperature: 10°C to 40°C

Relative humidity: maximum 65 %

Placement



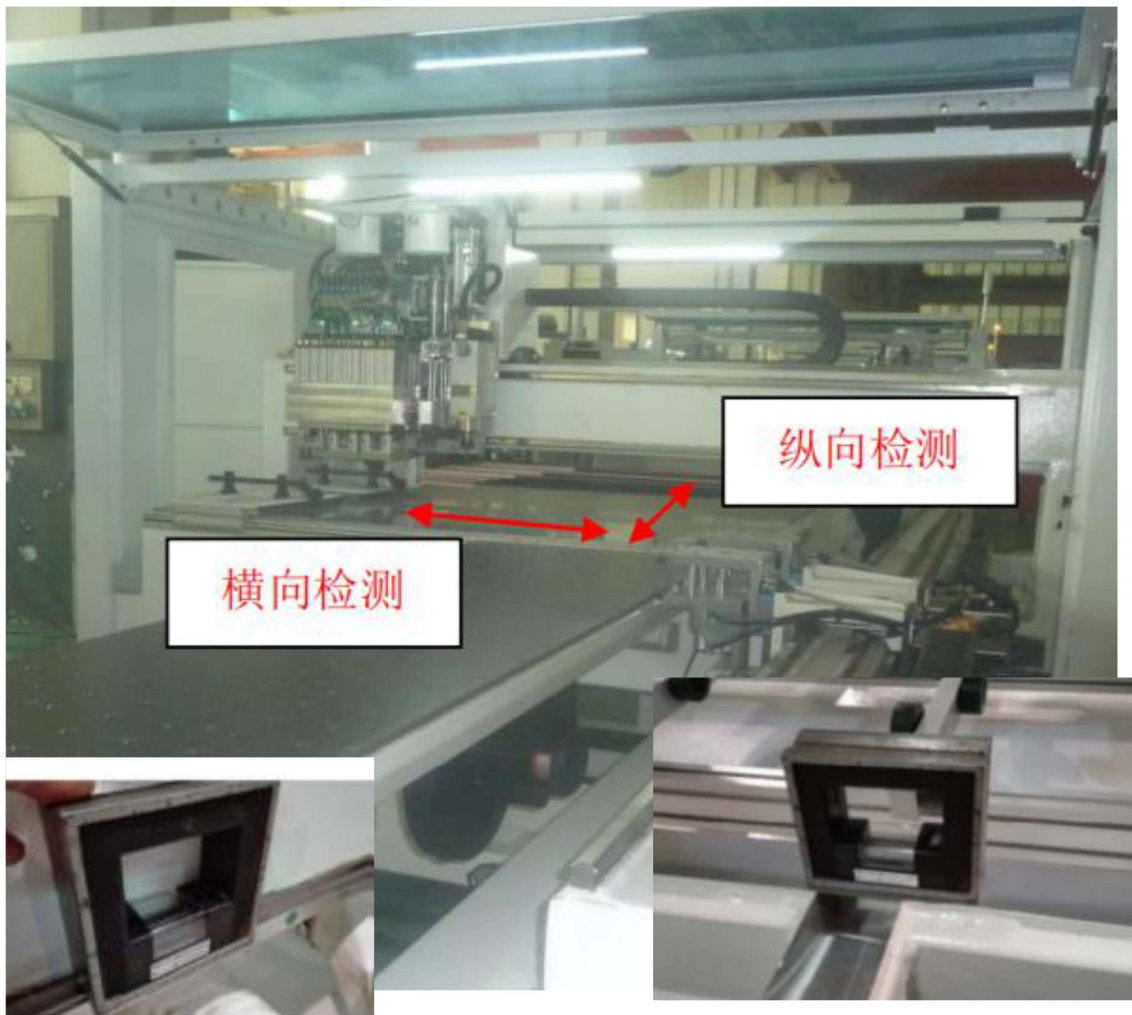


Horizontal adjustment

Use a professional level (accuracy: 0.05mm/m), and a plate with constant thickness ($\pm 0.01\text{mm}$).

Check the level by placing the board on the machine's workbench, and then placing the level again. Check for levelness in at least two vertical directions (X and Y) The levelness of the machine is adjusted by adjusting the anchor bolts of the machine.

Always measure on a machined surface, otherwise it won't be accurate.

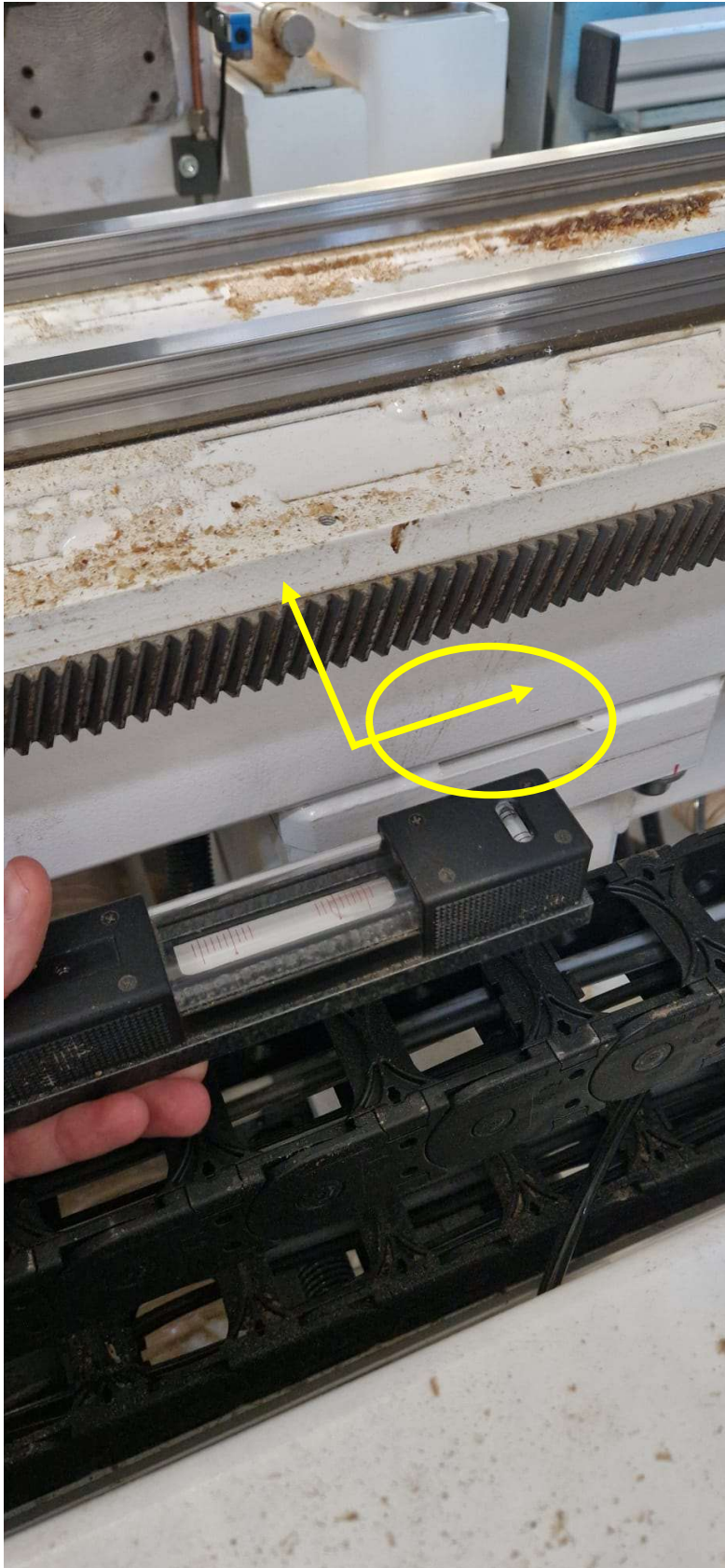






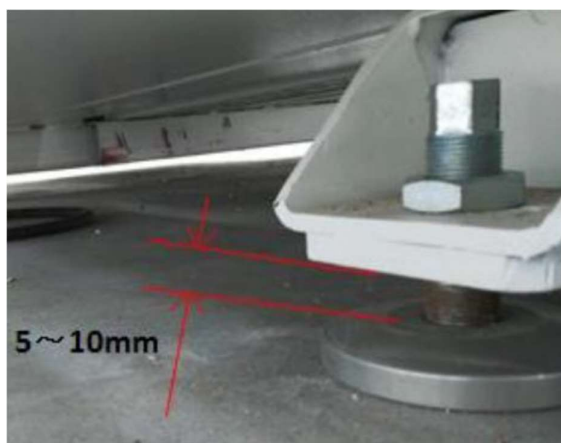








Adjustment, when necessary, needs to be done through the machine shoes, that is provided with the machine, adjust and keep measuring until it is aligned in X and Y.



Assembly

Guiderail

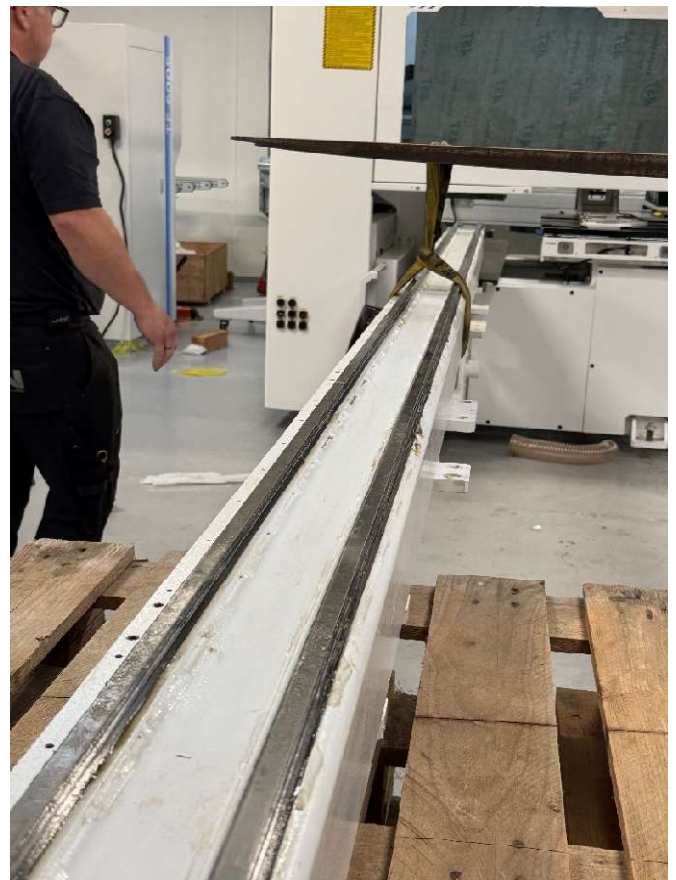
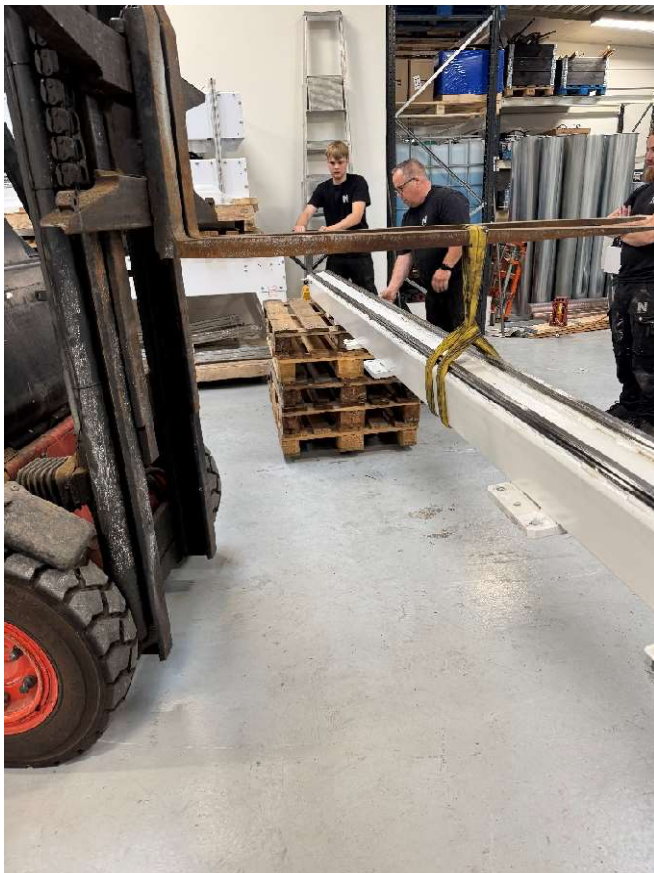
Description of assembly method and pictures of how to do so.

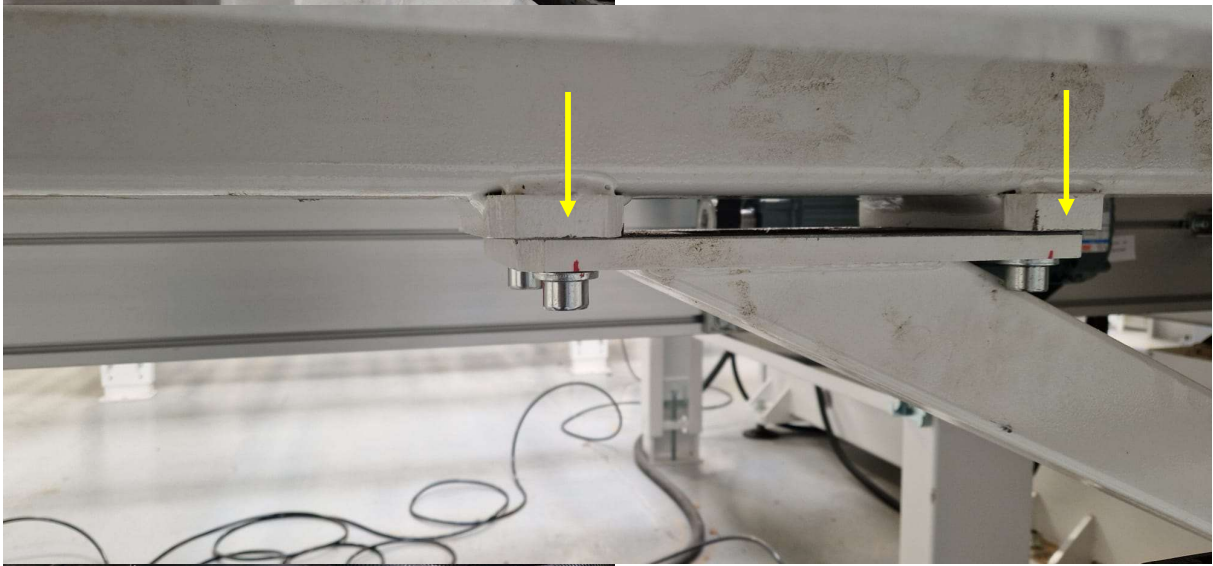
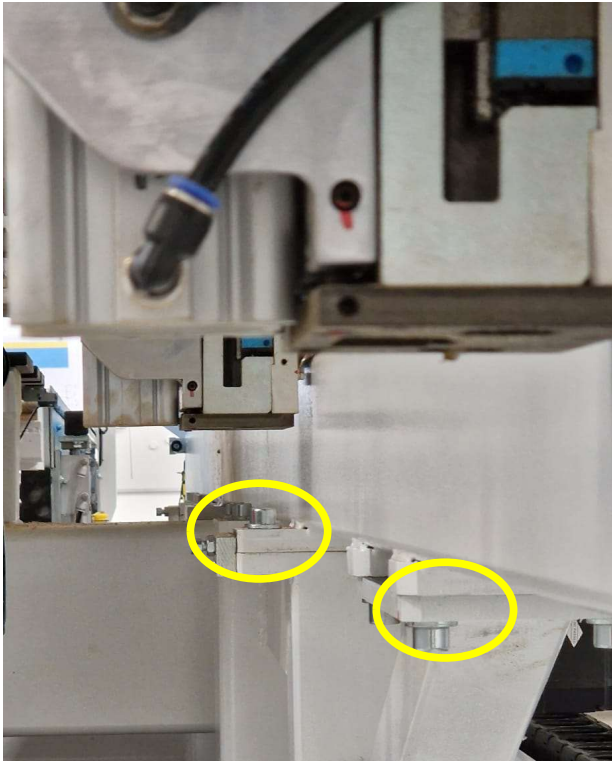
Align the machine as above description.

After that, positioning the guiderail for the clamps into the machine, needs to be placed on the 4 machined surfaces, that is the position holders for the guiderail.

Lift it by using a forklift, and place it on the apx. Same height as the surface of the position and roll it in, by using a pallet lifter, otherwise if available use a sideloader forklift.

Place It on the yellow marked machined surfaces.



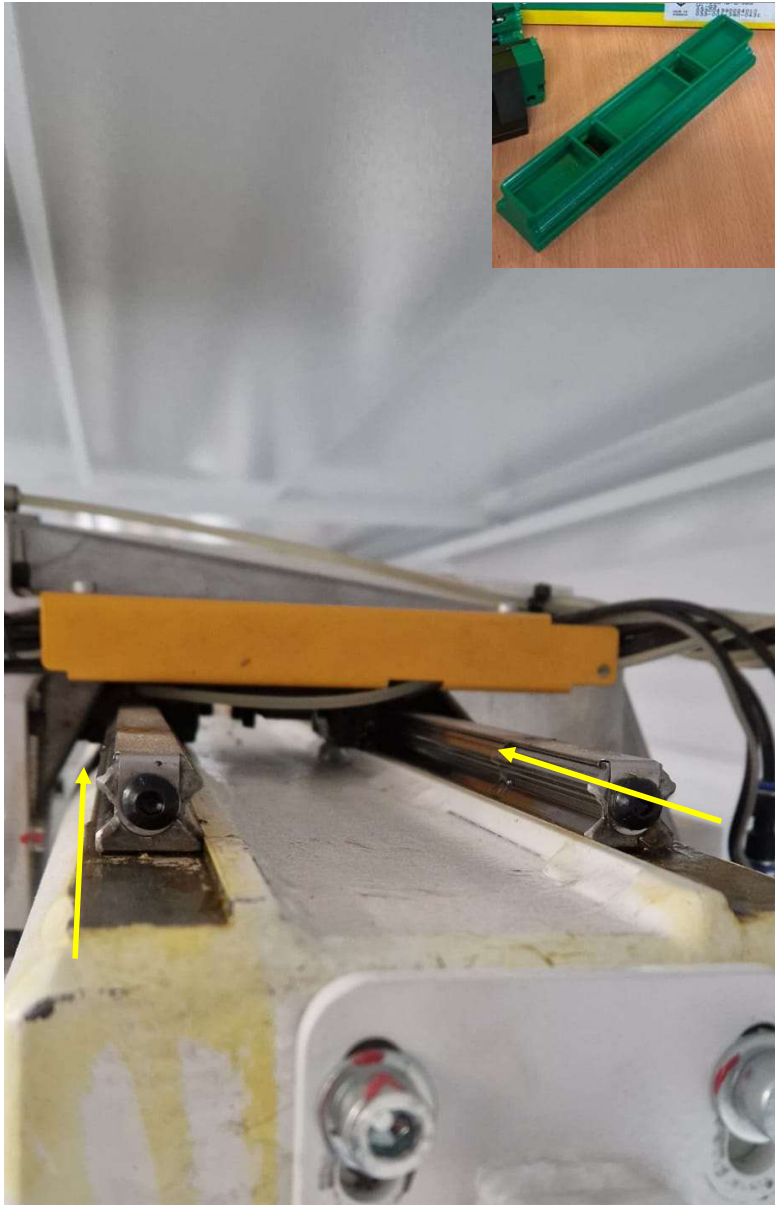


Level out the guiderail, again after the placement and bolting it down to the surface.



Clamping

Insert the clamping grippers by using the guide that is provided with the machine, for allowing the guides to slide over on the guiderail on the machine and to avoid bearings to drop out of the guide, remember to insert both grippers on the guide.

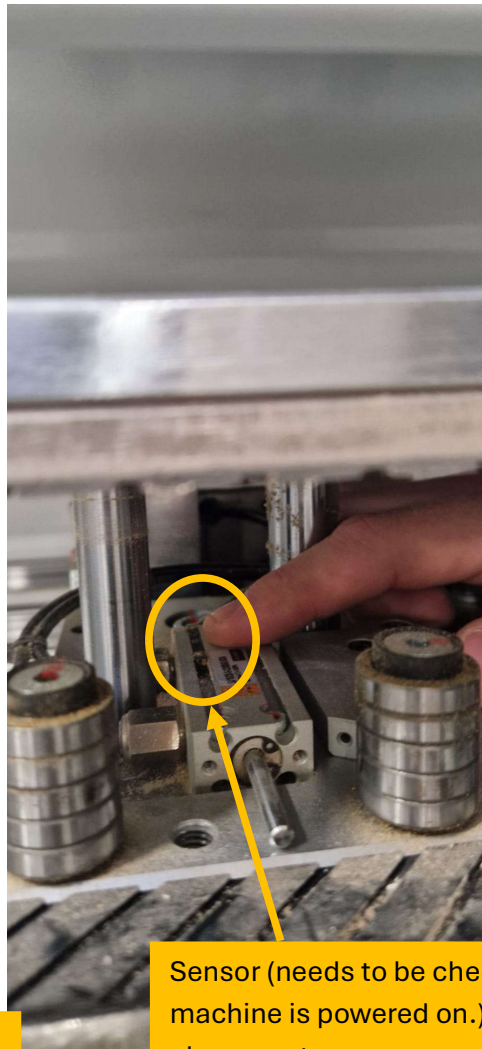




After inserting the grippers, connect all the wires and air connection to the clamps, and check that all connections are connected, all the way around.

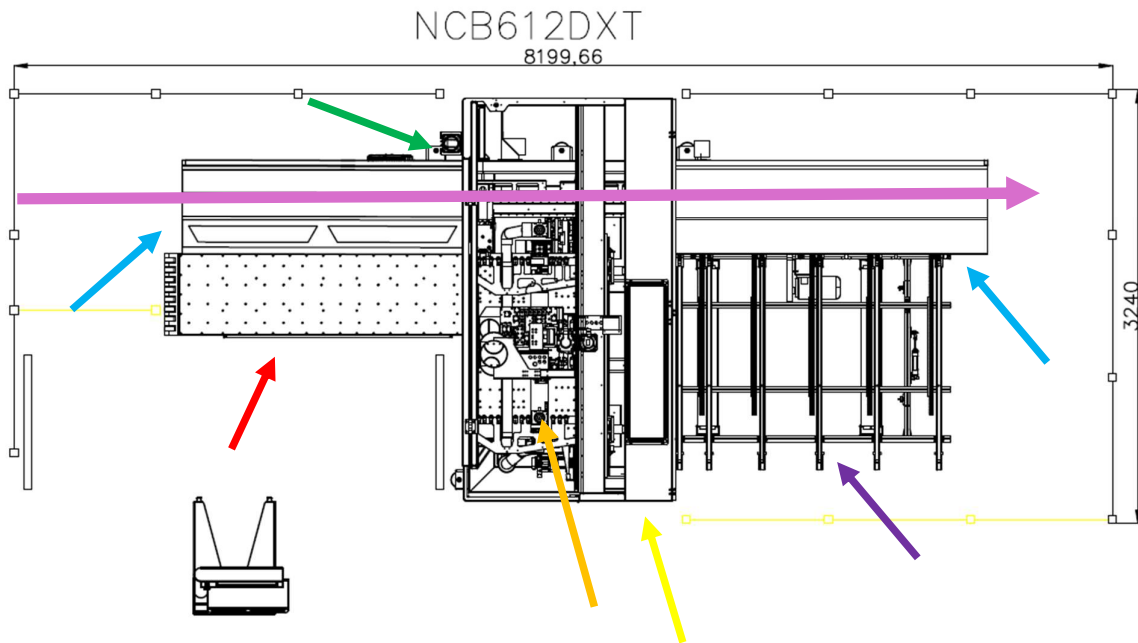


Air connection for opening and closing of the clamps.



Sensor (needs to be checked when machine is powered on.) this is apx. placement.

Assembly (Manuel machine)



- Assembly the shields, both sides. (blue)
- Assembly the Air floating table. (Red)
- Assembly the rear outfeeding table. (Purple)
- Assembly the fan motor on the back of the front table. (Green)
- Assembly the chip conveyor (option) if installed. (Orange)
- Assembly the rail for the energy chain. (Pink)
- Assembly the air condition (Yellow)

Remember to level out each component when they are mounted and bolted the machine, no need to use a machine level can be done with a normal leveller.

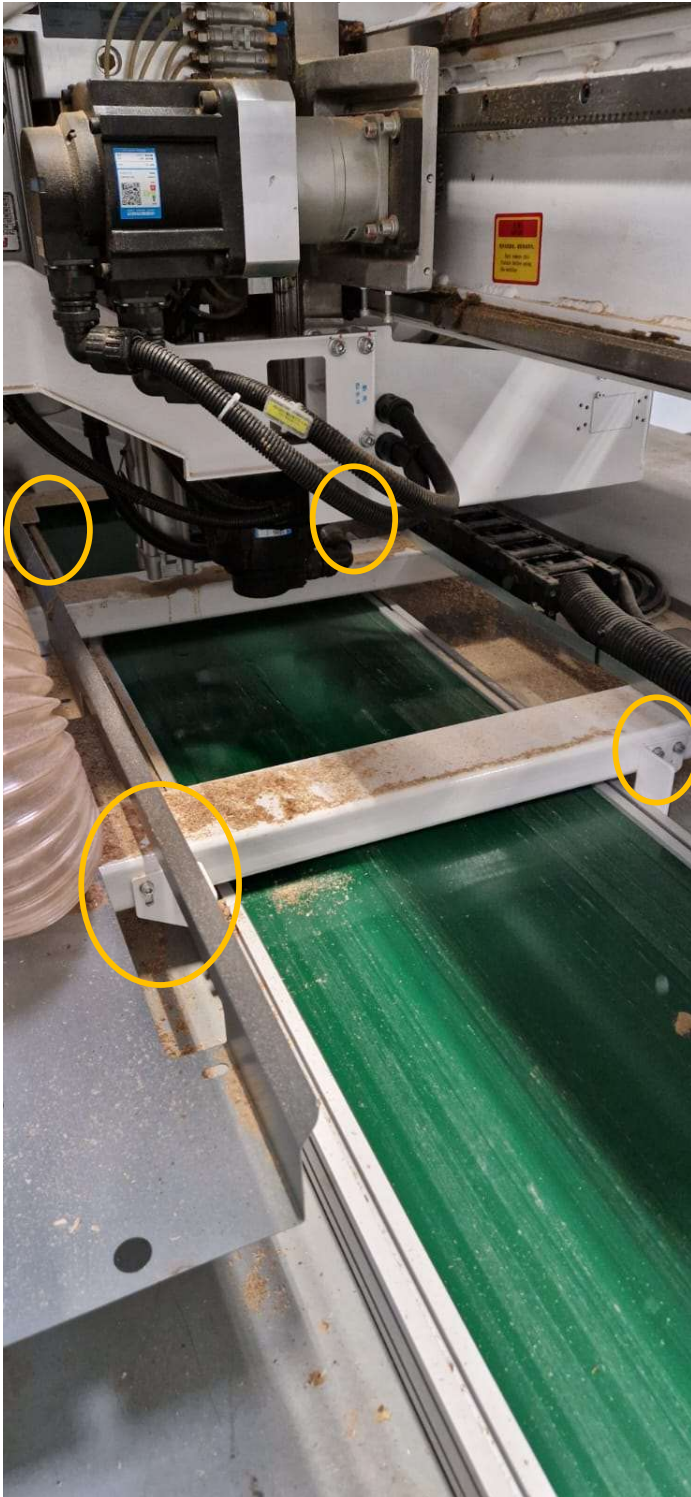




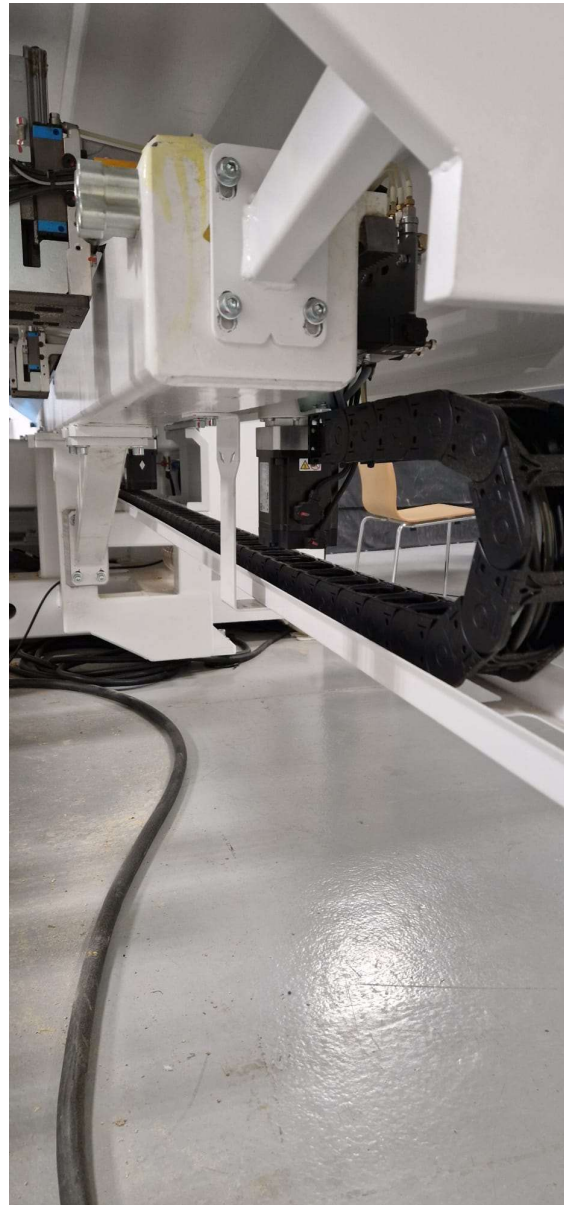
Mount the air fan, for the floating table.



The conveyor belt, needs to be mounted with these brackets, all 4 positions.



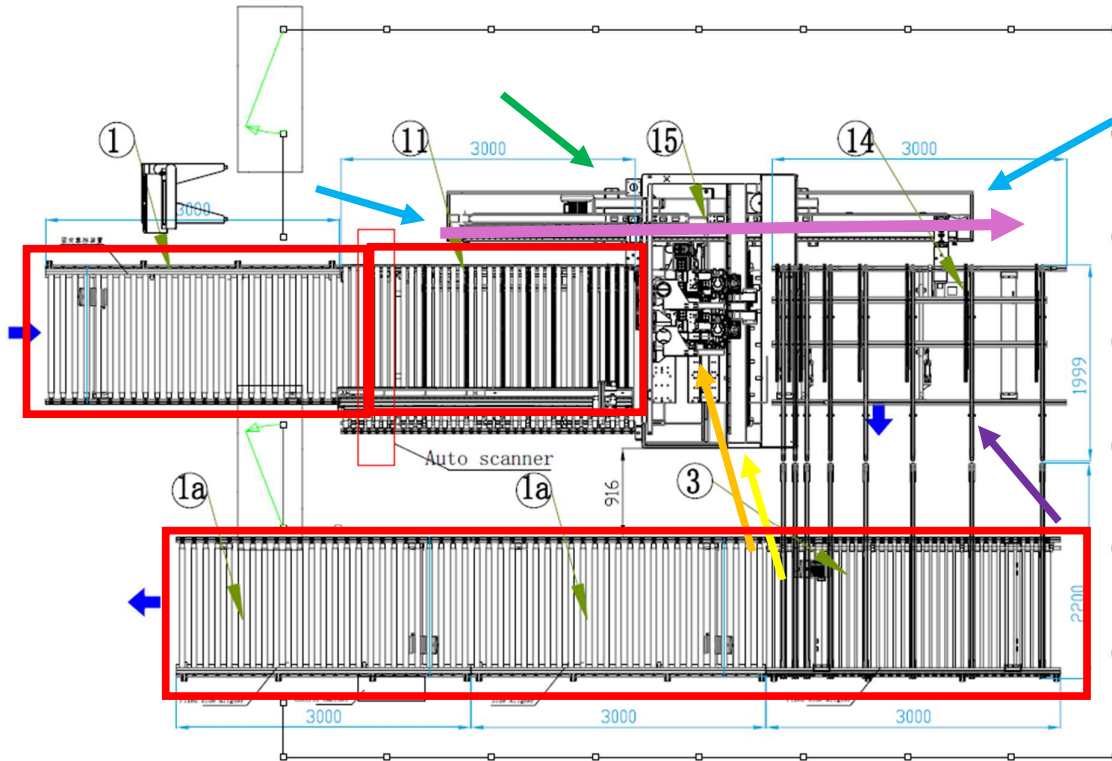
Assemble the energy chain holder, that's goes all the way underneath the guiderail.



The air condition unit, which is mounted on the side of the machine, needs to be connected to the machine afterwards.



Assembly [Automated machine]

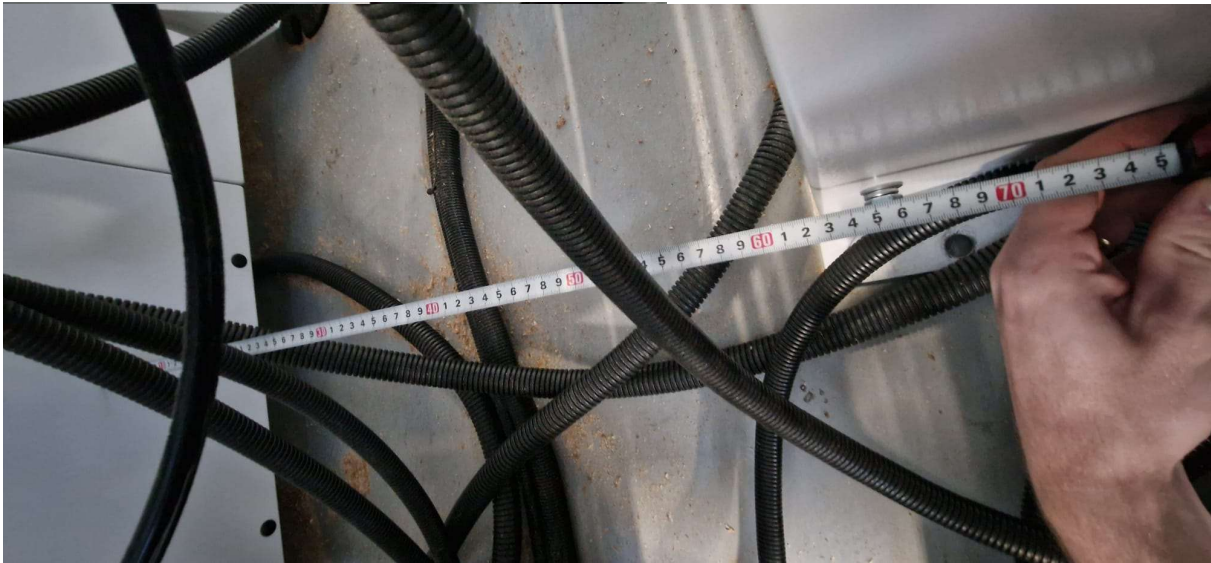


Assembly the machine as the manual model, the only difference is the conveyor systems that are attached to the machine.

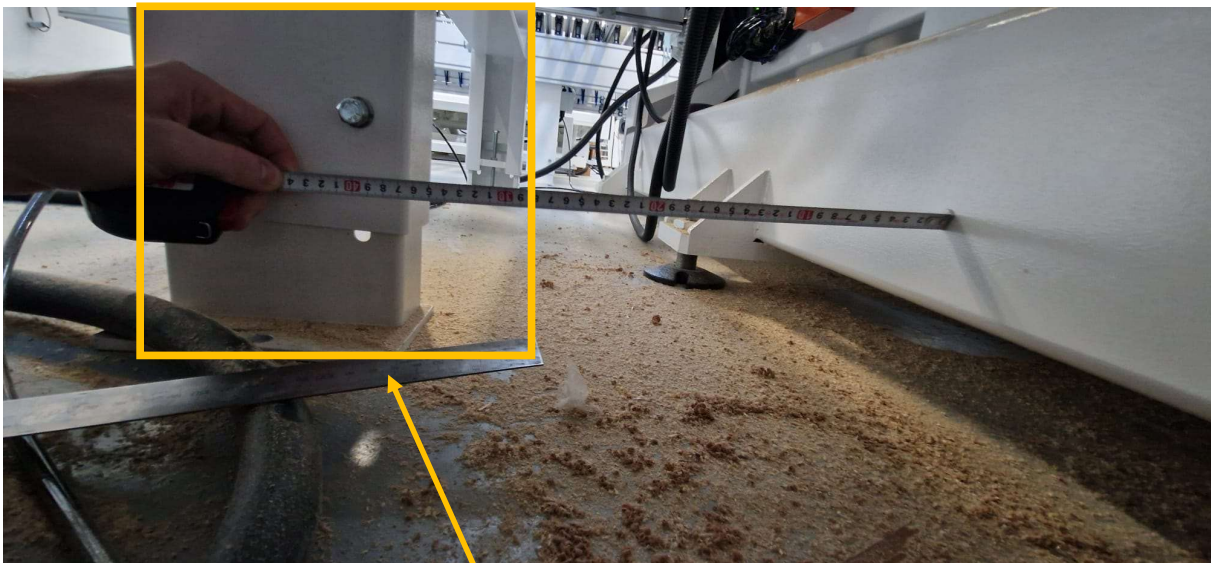
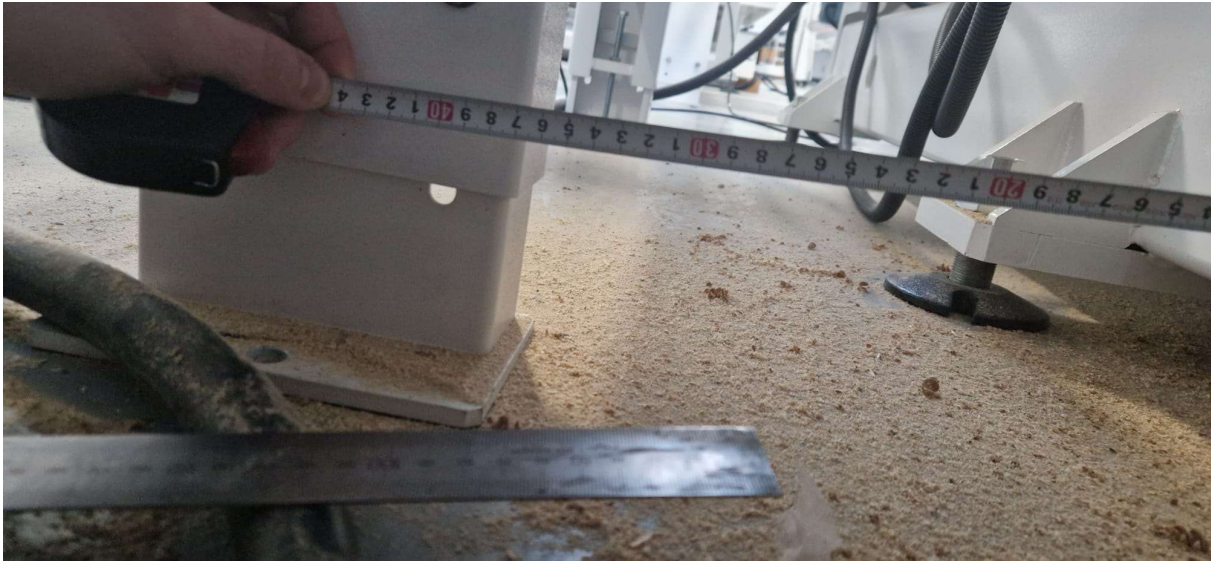
- Assembly the shields, both sides. [Blue]
- Assembly the rear outfeeding table. [Purple]
- Assembly the fan motor on the back of the front table. [Green]
- Assembly the chip conveyor (option) if installed. [Orange]
- Assembly the rail for the energy chain. [Pink]
- Assembly the air condition [Yellow]
- Assembly the roller conveyors [Red]



Distance from the front infeed table, measured from the legs. (630cm)

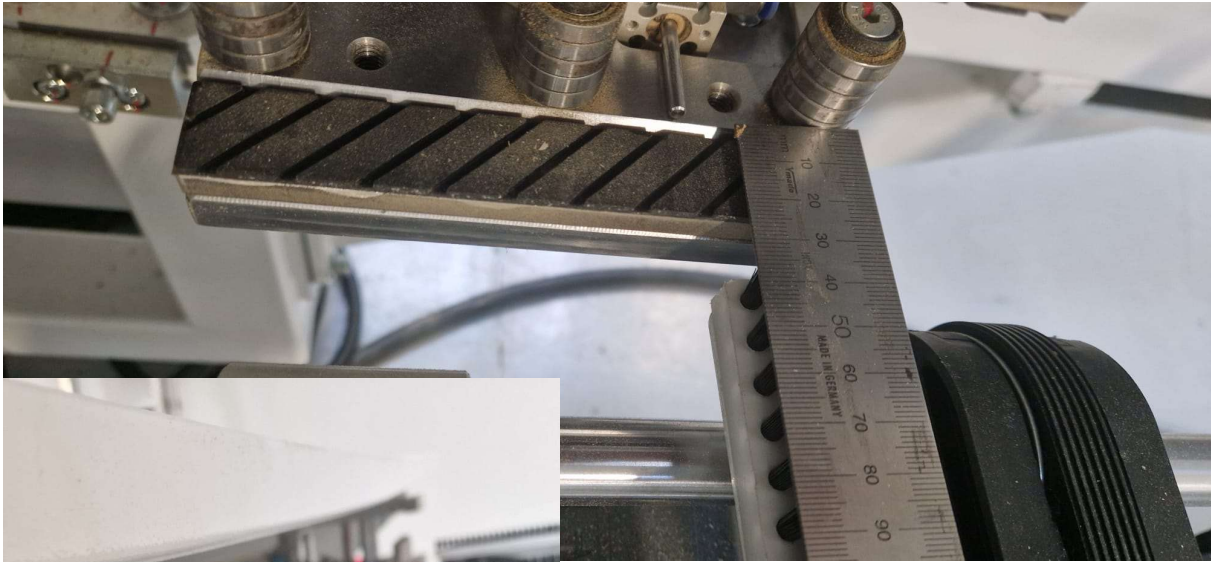


Apx. Distance from the legs of the rear outfeed table. (370 cm.)



To adjust the height of the tables,
adjust the 4 legs, that are placed
underneath the tables.

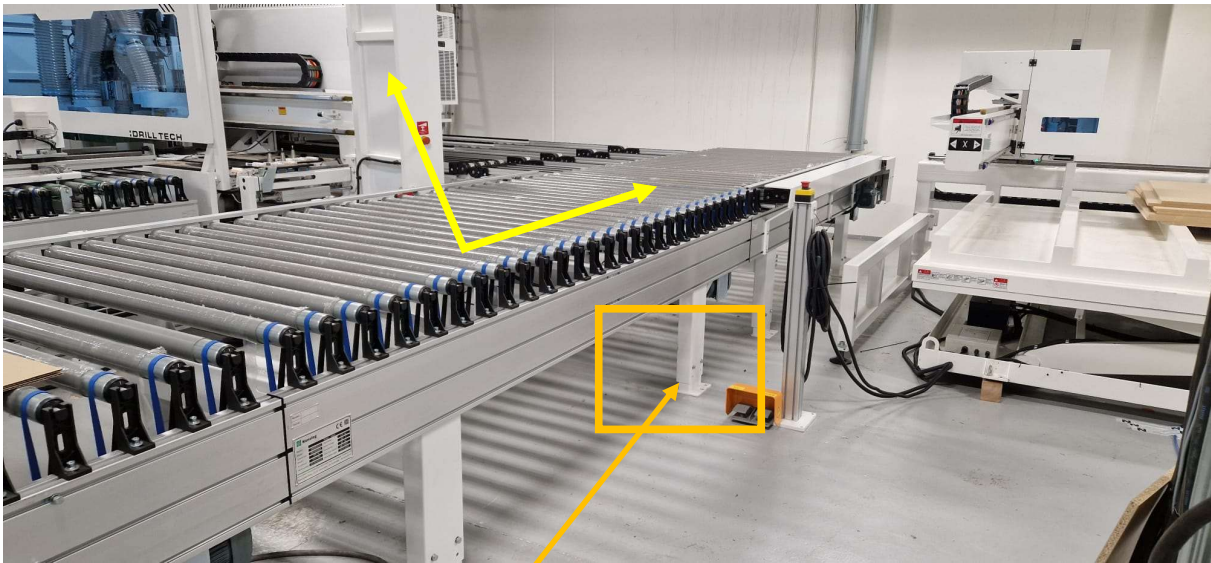
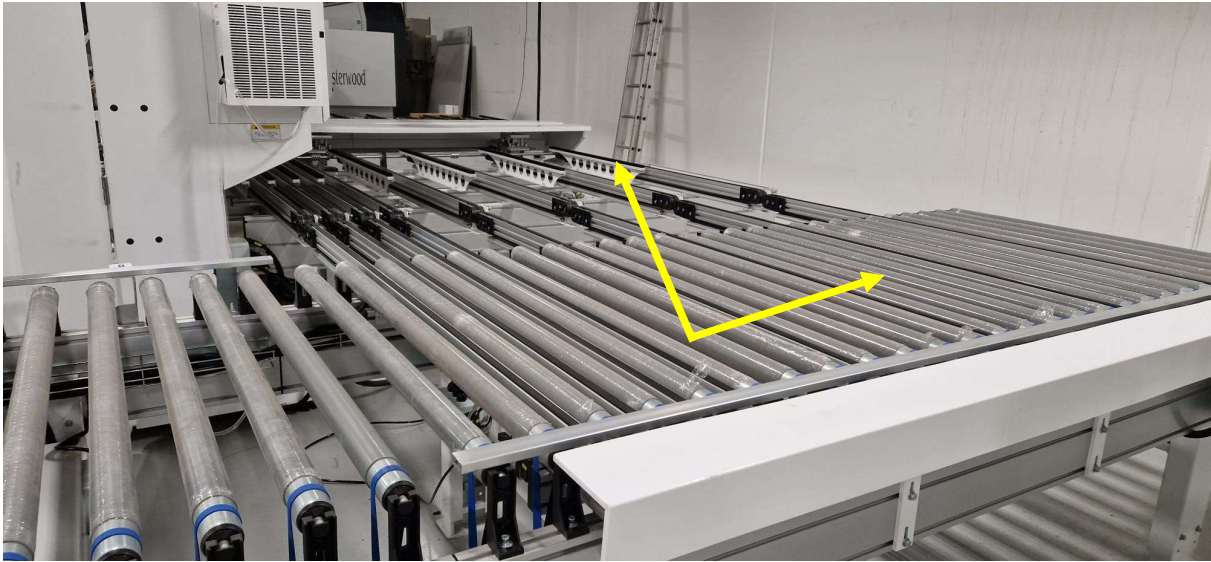
Distance from clamps to the rear outfeed table **(be careful when setting up these tables that close to the table, you can get collisions, that can cause a lot of damage!)**



Setup the tables, front and rear as close as you can by measurement and then run with the clamps manually through the machine and check the distance!

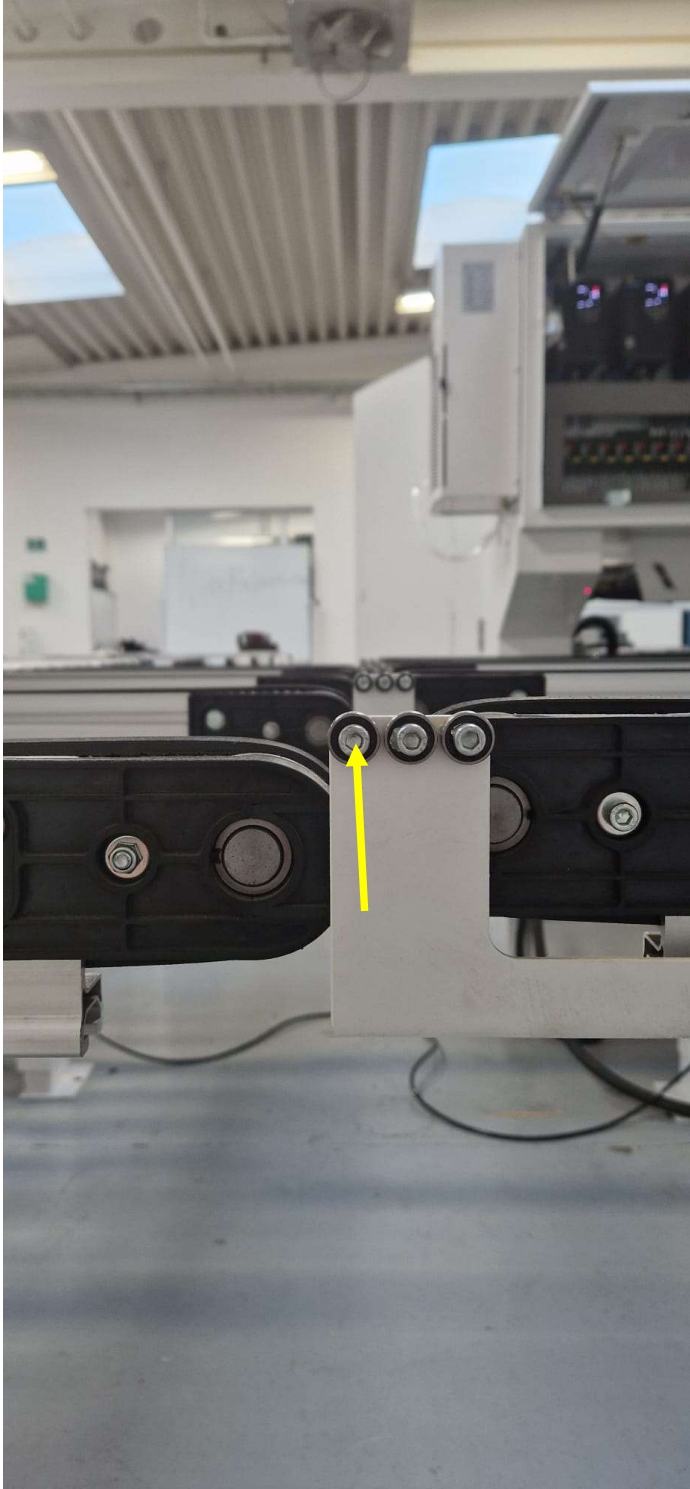


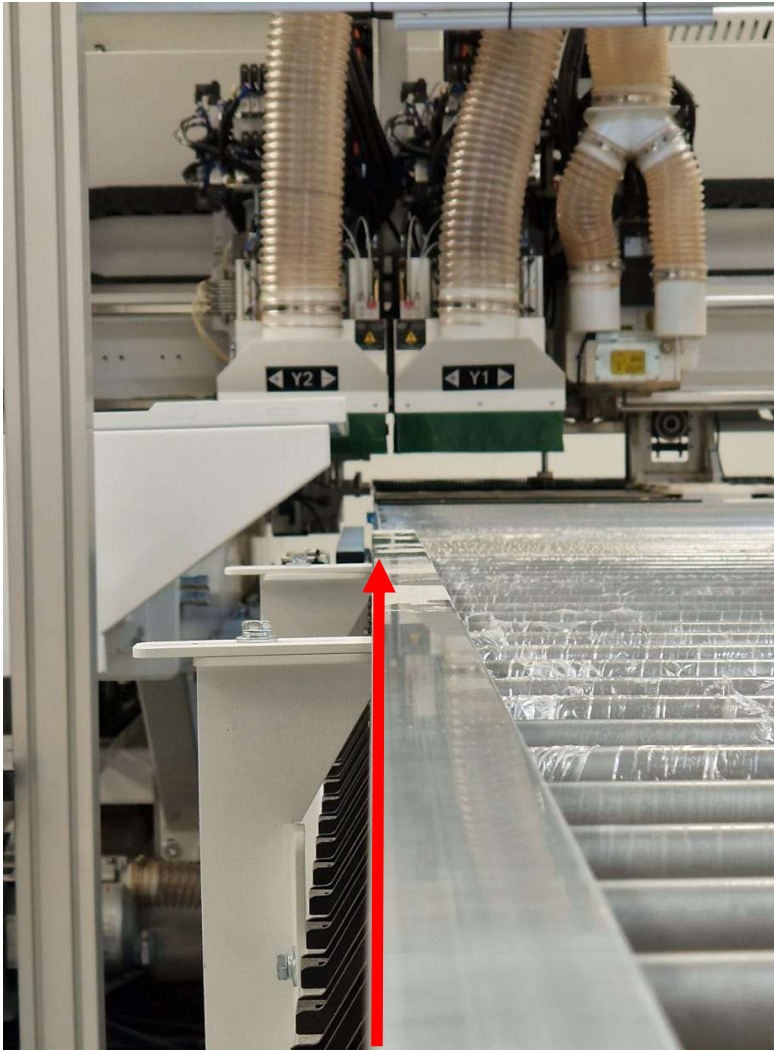
All tables must be leveled out in X and Y.



To adjust the height of the tables, adjust the 4 legs, that are placed underneath the tables.

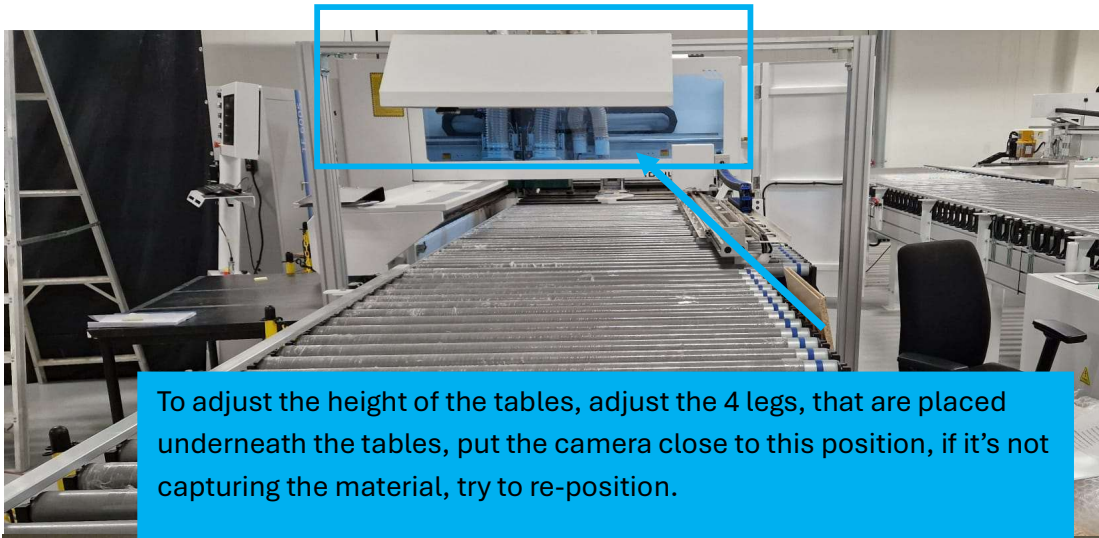
We have placed the rear outfeed table a little higher, so we avoid collision or faulty push out of materiale from the rear.



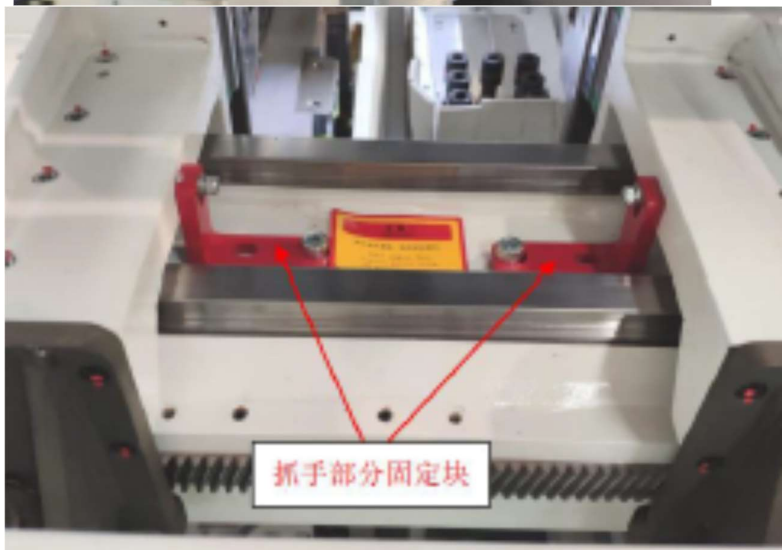


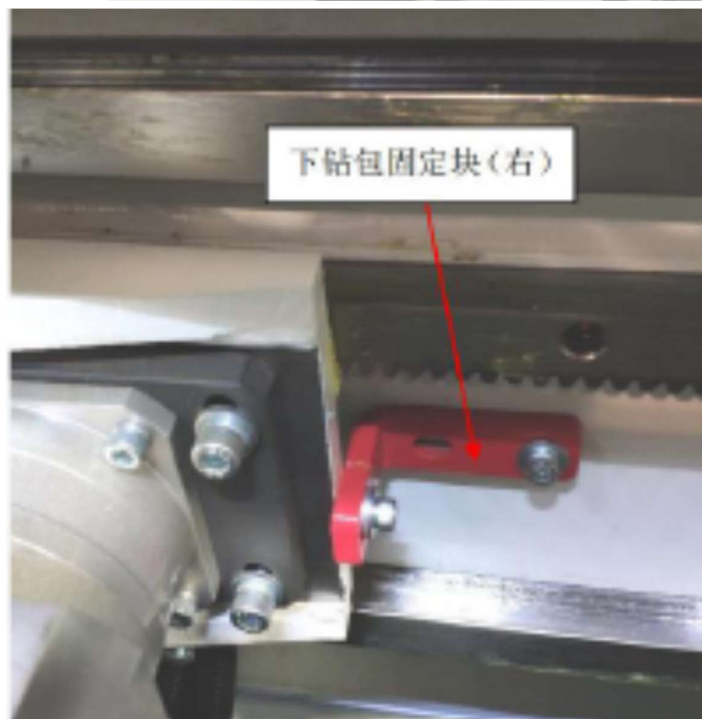
The infeed table's alignment ruler, which is located on the left side.

Needs to be aligned by using the backside of the alignment ruler, otherwise the position of the camera, will not work for the smaller piece's



Removal of transportation brakes / holders.





Always double check, that all stops are removed from the machine before running!

Power Connection

The wiring steps are as follows:

1. Set the main switch in the area where the equipment is installed at "OFF".
2. Set the main switch of the equipment at "OFF". And then open the power leading-in box of the machine.
3. Connect one end of the cable to the power leading-in box of the machine. Insert the three-phase wires of the cable into the corresponding holes L1, L2, L3 in the power inlet box of the machine; Then, insert the cable ground wire (yellow/green) into the ground wire hole PE (yellow/green) in the machine power inlet box.
4. Connect the other end of the cable to the user's power supply; follow the corresponding phases L1, L2, L3, and ground wire PE.

The equipment is equipped with a phase sequence protector, the phase sequence is normal, and the green light is on.

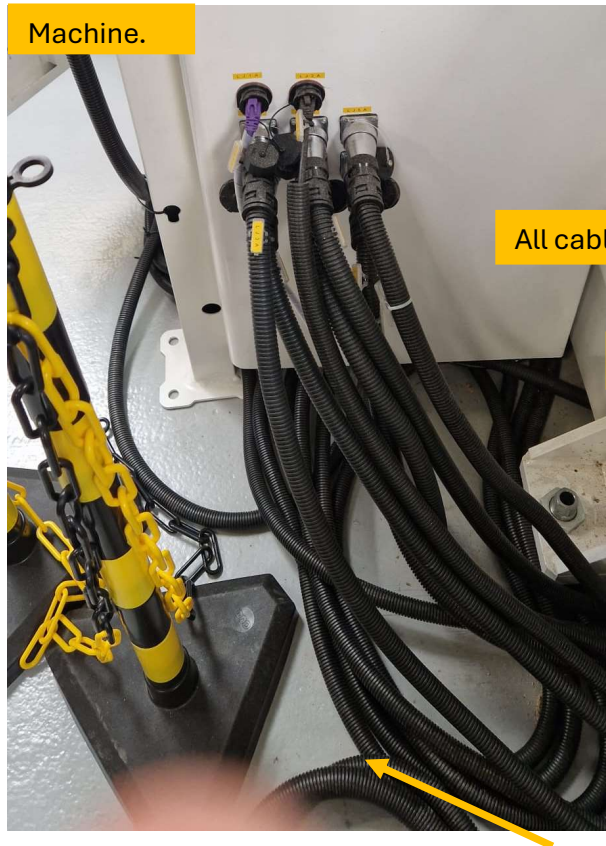


Phase sequence status		Indicator
Display	Meaning	lamp
NORMAL	Normal phase sequence	Green light
REVERSE	Reverse phase sequence	Yellow light
SCARCITY	Phase loss and missing connection	Red light



For safety reason, always make the "ground" connection a bit longer!

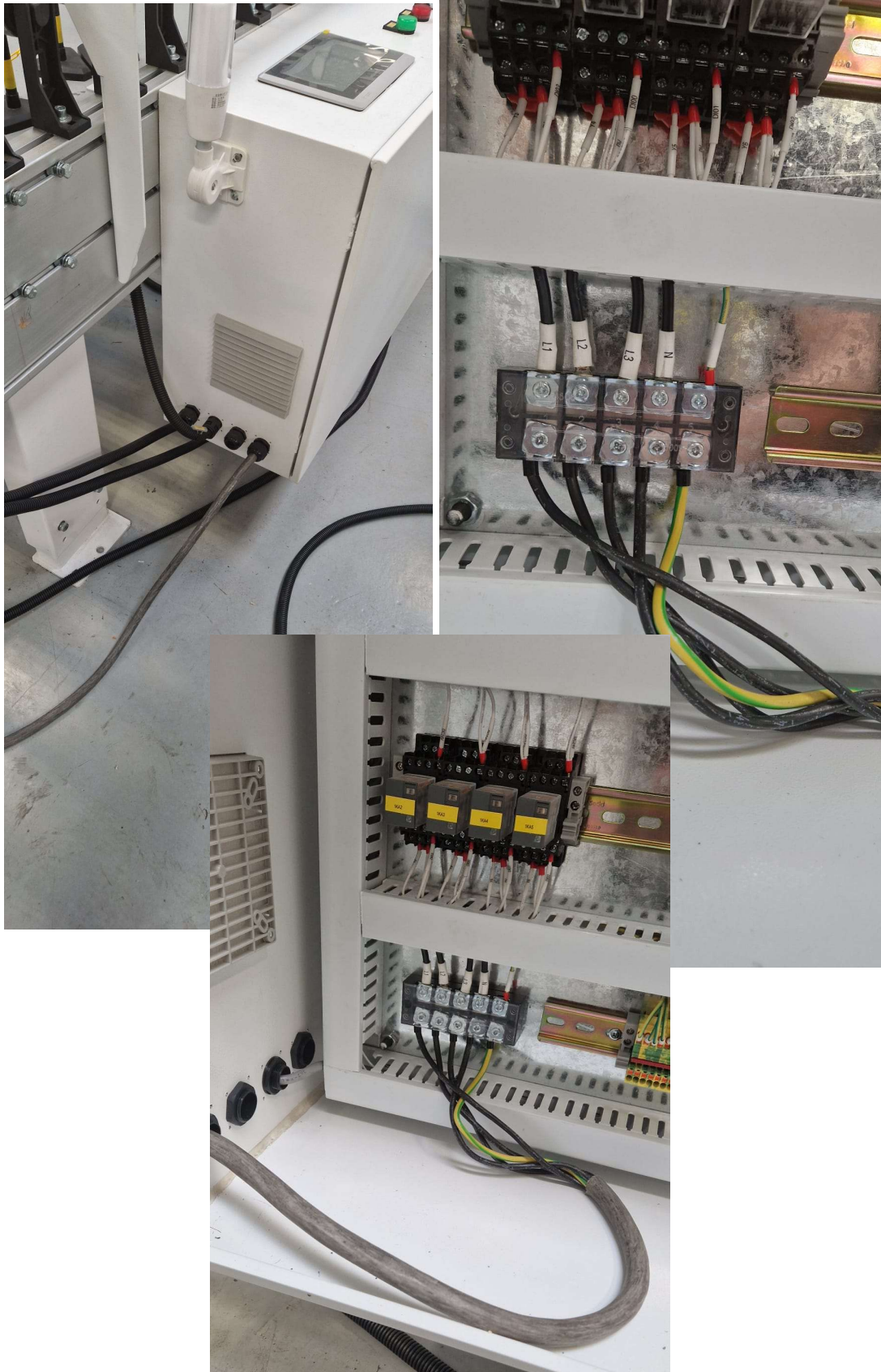
Assemble all connections, depending on the model, that you have received, the amount of cables and connections can change **(see section with Power connection and air connection.)**



All cables are labeled



If you have the automated line, you need to connect another power supply for the external conveyor unit, because this is controlled by it self and not by the machine.

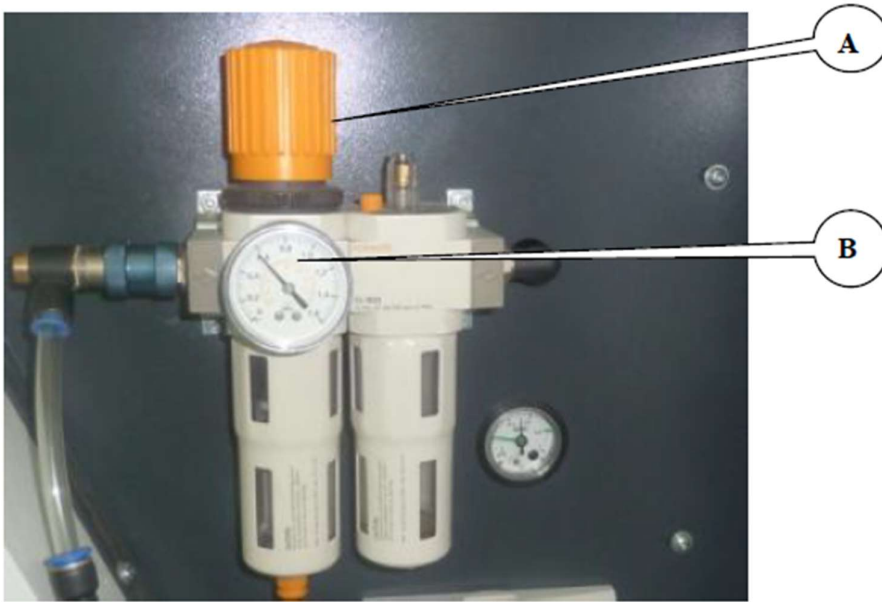


Connection of Air circuit

Connect the gas circuit air supply through the pressure regulating filter device (FRL).

The steps are as follows:

1. Pull out the knob "A" of the pressure regulating filter unit and turn the knob until the pressure gauge "B" displays a pressure of 0.6 MPa - 0.7 MPa.
2. After adjusting the force, press the knob "A" again.



Remember to connect air to the external unit's if you have the automated line.



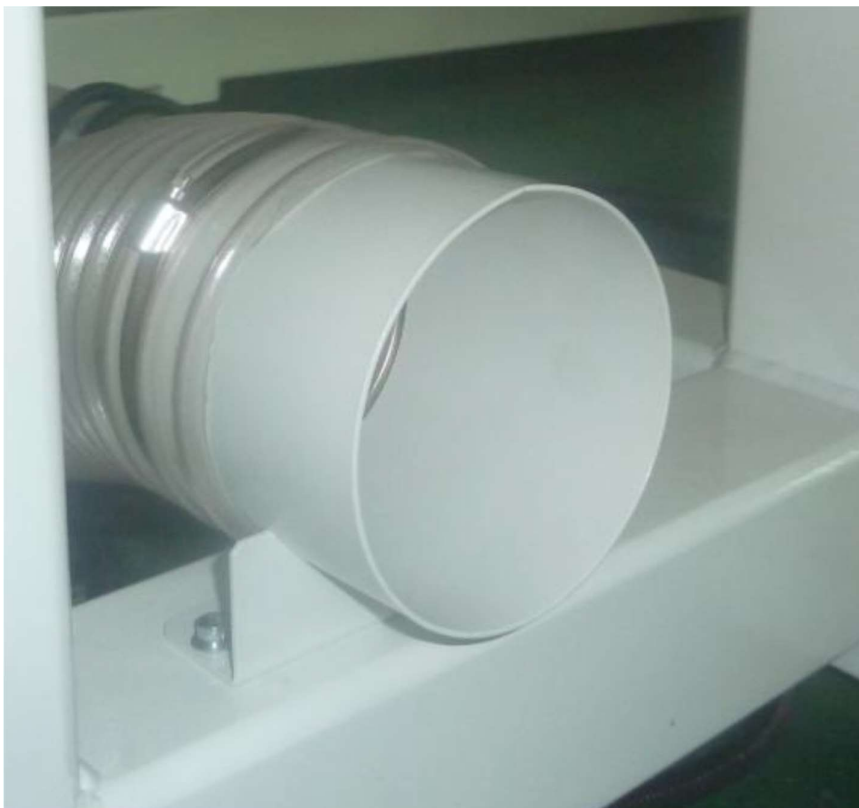
Located here, below the rollers.



Connection of dust collection opening

The diameters of the upper and lower suction ports are $\phi 150\text{mm}$ and $\phi 125\text{mm}$ respectively.

Connect with a hose and fasten it with a pipe strap.









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